SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Voce Qns

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 – Viva Voce Qns**	**Assessment:**	**Assessment Tool 3 – Viva Voce Qns**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	Y. Jayesh Ranjan	**Reg. No.:**	192472201
Section / Batch:	CSE – AI	**Date:**	25/08/26

Code of Conduct

I Y. Jayesh Ranjan – 192472201 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Jayesh Ranjan

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

A. Introduction to Statistical Inference

1. What is statistical inference?

Statistical inference is the process of using sample data to make conclusions about a population.

It helps us estimate population values and make decisions based on data.

2. Why is statistical inference important in data science?

It helps data scientists draw conclusions about large populations using limited data.

It is useful for prediction, decision-making, and testing assumptions.

3. What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics summarizes and presents the data that we have collected.

Inferential statistics uses sample data to make conclusions about a larger population.

4. What is a population in statistics?

A population is the complete set of individuals or observations that we are interested in studying.

For example, all students in a college can be considered a population.

5. What is a sample?

A sample is a smaller group selected from a population for analysis.

For example, 100 students selected from a college are a sample.

6. Why do data scientists usually work with samples instead of complete populations?

Studying the entire population can be expensive, time-consuming, or difficult.

A carefully selected sample can provide useful information about the whole population.

7. What is a population parameter?

A population parameter is a numerical value that describes a characteristic of the entire population.

Examples include population mean, population proportion, and population variance.

8. What is a sample statistic?

A sample statistic is a numerical value calculated from sample data.

Examples include sample mean, sample proportion, and sample variance.

9. Give a real-world example where statistical inference is used in industry.

A company can survey a sample of customers to estimate overall customer satisfaction.

The results from the sample can be used to make decisions for the entire customer population.

10. What are the two major activities involved in statistical inference?

The two major activities are estimation and hypothesis testing.

Estimation finds population values, while hypothesis testing helps make decisions about population claims.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

The frequentist approach uses sample data to make conclusions about fixed population parameters.

It focuses on the long-run behavior of repeated random samples.

12. What is the basic idea behind frequentist statistics?

The basic idea is that probability describes the long-run frequency of events.

Population parameters are treated as fixed, while the observed data can vary.

13. How does the frequentist approach differ from the Bayesian approach?

In the frequentist approach, population parameters are treated as fixed values.

In the Bayesian approach, parameters can be treated as uncertain and assigned probability distributions.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

The population parameter is considered fixed.

The sample data is considered random because different samples can produce different results.

15. What does long-run frequency mean in frequentist statistics?

Long-run frequency means the proportion of times an event occurs when an experiment is repeated many times.

It is the basic interpretation of probability in frequentist statistics.

16. Why is random sampling important in the frequentist approach?

Random sampling helps ensure that the sample represents the population fairly.

It also allows us to study the sampling variability of estimates.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

No, a population parameter is considered a fixed but unknown value.

The randomness comes from the sample data, not from the parameter.

18. Give an example of a frequentist problem in a data science application.

A data scientist may estimate the average spending of all customers using a random sample of customers.

The sample mean is used to make an inference about the fixed population mean.

C. Point Estimation

19. What is a point estimate?

A point estimate is a single numerical value used to estimate an unknown population parameter.

For example, the sample mean can be used as a point estimate of the population mean.

20. What is an estimator?

An estimator is a rule or formula used to calculate an estimate of a population parameter from sample data.

For example, the sample mean is an estimator of the population mean.

21. What is the difference between an estimator and an estimate?

An estimator is the formula or method used for estimation.

An estimate is the actual numerical value obtained after applying the estimator to a sample.

22. What is the point estimator for the population mean?

The sample mean is commonly used as the point estimator for the population mean .

It is calculated by adding all sample values and dividing by the sample size.

23. How do you estimate a population proportion from a sample?

The population proportion is estimated using the sample proportion.

It is calculated as the number of successes divided by the total sample size.

24. What is the point estimate of population variance?

The sample variance is used as the point estimate of population variance.

It measures how much the observations in the sample vary around the sample mean.

25. Why is the sample mean commonly used to estimate the population mean?

The sample mean is an unbiased estimator of the population mean.

It is also simple to calculate and becomes more reliable as the sample size increases.

26. What properties should a good estimator have?

A good estimator should ideally be unbiased, consistent, and have low variance.

These properties make the estimator reliable and close to the true population parameter.

27. What is an unbiased estimator?

An unbiased estimator is an estimator whose expected value is equal to the true population parameter.

It does not systematically overestimate or underestimate the parameter.

28. What is meant by the bias of an estimator?
Bias is the difference between the expected value of an estimator and the true population parameter.
An estimator with zero bias is called an unbiased estimator.
29. What is the difference between bias and variance of an estimator?
Bias measures the systematic difference between the estimator and the true parameter.
Variance measures how much the estimates change from one sample to another.
30. Why can two different samples produce different point estimates for the same population parameter?
Different samples contain different observations because of random sampling.
Therefore, their calculated statistics, such as sample means, can also be different.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?
A point estimate is calculated from a sample, and different samples can give different estimates.
Therefore, the estimate may not be exactly equal to the true population parameter.
32. What is the sampling distribution of an estimator?
The sampling distribution is the probability distribution of an estimator obtained from many possible samples.
It shows how the estimator varies from sample to sample.
33. What is the standard error?
Standard error measures the variability of a sample statistic or estimator across repeated samples.
A smaller standard error means the estimate is more precise.
34. How is standard error different from standard deviation?
Standard deviation measures the spread of individual observations in a dataset.
Standard error measures the spread or variability of a sample statistic across samples.
35. How does increasing the sample size affect the standard error of the sample mean?
Increasing the sample size decreases the standard error of the sample mean.
The standard error is proportional to , so larger samples give more precise estimates.
36. Why does variability in an estimate decrease when the sample size increases?
A larger sample contains more information about the population.
Therefore, random fluctuations have less effect on the estimated value.

37. What is the relationship between standard error and confidence intervals?

Standard error is used to determine the margin of error in a confidence interval.

A smaller standard error generally produces a narrower confidence interval.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect?

No, different sample means are normal because samples contain different observations.

The difference can occur due to natural sampling variability.

E. Confidence Intervals

39. What is a confidence interval?

A confidence interval is a range of values used to estimate an unknown population parameter.

It gives an interval estimate instead of providing only a single point estimate.

40. Why is a confidence interval more informative than a point estimate?

A point estimate gives only one value and does not show its uncertainty.

A confidence interval gives a range and indicates the possible uncertainty in the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

It means that if we repeatedly take samples and construct intervals using the same method, about 95% of those intervals will contain the true parameter.

It does not mean there is a 95% probability that one fixed interval contains the parameter.

42. What is the difference between a confidence level and a confidence interval?

The confidence level is the percentage of confidence associated with the estimation method, such as 95%.

The confidence interval is the actual range calculated from the sample data.

43. What factors determine the width of a confidence interval?

The width depends mainly on the confidence level, sample size, and variability in the data.

Higher variability or higher confidence generally makes the interval wider.

44. How does increasing the sample size affect the width of a confidence interval?

Increasing the sample size generally makes the confidence interval narrower.

This happens because a larger sample reduces the standard error.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

Increasing the confidence level makes the confidence interval wider.

A wider interval is needed to provide a higher level of confidence.

46. What is the difference between a z-confidence interval and a t-confidence interval?

A z-confidence interval uses the normal distribution, while a t-confidence interval uses the t-distribution.

The t-distribution is especially useful when the population standard deviation is unknown.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

The t-distribution is used when the population standard deviation is unknown and is estimated using the sample standard deviation.

It is especially important for smaller sample sizes.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Hypothesis testing is a statistical method used to evaluate a claim about a population.

It helps us decide whether the sample data provides enough evidence against the claim.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

The null hypothesis (H_0) represents the default or existing claim being tested.

The alternative hypothesis (H_1) represents the claim or difference we want to find evidence for.

50. How can a confidence interval be used to make a decision about a hypothesis?

We compare the hypothesized parameter value with the confidence interval.

If the hypothesized value falls outside the interval, we reject the null hypothesis at the corresponding significance level.

Answers:

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